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First-Round Questions to the Candidate

Paper 4

Selection Functionals on Path Space for Stochastic Processes Without a Generator: Finite-Grid Gibbs Selection and Self-Normalized Estimation

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Identifiers:	Q-P04-01 to Q-P04-16 (technical report); Q-PF-06, Q-PF-09 (portfolio-level evaluation)
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This document belongs to the examination record (03_examiner_questions). It lists the questions put to the candidate in the first round of the examination of Paper 4 and gives each a stable identifier, by which the Report, the Technical Annex and the correspondence can cite it.

1 Sources and identifiers

Technical report. Section 2 transcribes Section 7, "Questions to the author", of the Doctoral Examiner's technical report on Paper 4, `TR-P04-DOCv1-EX01-Rv1.pdf`. The report identifies the version examined as SSRN, DOI 10.2139/ssrn.6922758, 26 July 2026, and carries the evaluation date 4 September 2026. The Doctoral Examiner sent it to the candidate on 14 September 2026 (COR-P04-02), and the candidate answered, to both examiners, on 16 September 2026 (COR-P04-03).

Identifiers. `Q-P04-`k is Question k of that report. The candidate's letters and the Technical Annex cite the questions by these numbers. The text is the report's, word for word; its section signs (§) refer to sections of the paper, and its finding numbers to findings of the report. Notes in small type are not part of the questions. They identify unresolved references, later corrections and related questions.

Independent evaluation. The independent evaluator's review of Paper 4 (`TR-P04-DOCv1-EX02-Rv1.pdf`, Exhibit J, 02_technical_evaluations) states its points as comments and poses no questions. It was sent to the candidate on 11 September 2026 and is filed as COR-P04-01; the candidate answered it together with the technical report on 16 September 2026 (COR-P04-03). It is not transcribed here.

Portfolio-level evaluation. Section 3 transcribes the questions of the portfolio-level technical evaluation (version 2, 4 September 2026; Exhibit H, Section 7, 02_technical_evaluations) that concern Paper 4. `Q-PF-`k is Question k of that evaluation and has the same identifier in all six documents; Q-PF-09 concerns all six papers and appears in each. The evaluation names the works W1–W6, and Wn is Paper n.

2 Questions of the technical report

Source: `TR-P04-DOCv1-EX01-Rv1.pdf`, Section 7, Questions 1–16.

Q-P04-01 Show that $e^{-\lambda S_{\nu,M}}$ is proportional to the surrogate's Euler path density raised to λ for constant-divergence surrogates, and state the consequence $Q_{\nu,M} \propto \Pi_M\, p_\nu^\lambda$. Why multiply the two densities rather than divide, which is what every reweighting method in the adjacent literature does?

Q-P04-02 Derive (1) for the Gaussian case and say what it implies for the quadratic variation of the selected measure relative to the proposal.

Note. (1) is the display of the report, Section 3.2, for Gaussian proposal and surrogate: $1/\sigma_{\text{eff}}^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2$. The report states it as exact. It holds to leading order in the grid step, and the exact interior-row relation is in k^2/σ^2. The candidate corrected it, and the Doctoral Examiner accepted the correction in his letter of 23 September 2026 (COR-P04-06; Technical Annex, Section T.2(i)).

Q-P04-03 Construct an explicit example in which a divisible surrogate family is bounded away from an indivisible target in kernel norm while the selected measure equals the true grid law exactly. What does that show about the sentence opening §7.2, and what remains of the claim that the conditional structure is forced?

Q-P04-04 Given (1), for which proposals is hypothesis (D1) attainable at all? Is the tolerance $\eta_{\nu,M}$ controlled by any property of the target, or only by the proposal–surrogate pair?

Note. For (1), see the note to Q-P04-02.

Q-P04-05 Is there a temperature schedule λ_M under which the continuum limit is both non-degenerate and non-trivial? The evidence from your own example is that holding the chi-square divergence fixed drives the selection displacement to zero.

Q-P04-06 Precisely which continuous-path object fails in Proposition 2.3, and why does the argument not rule out a Girsanov change of path measure or an Onsager–Machlup small-ball functional?

Q-P04-07 Derive the divergence term of the classical action from a small-ball or Stratonovich formulation. In what sense is it, and in what sense is it not, an Itô correction?

Q-P04-08 Position the construction against Girsanov path reweighting of externally generated ensembles and against reweighting with discretised stochastic actions. The two distinctions offered in §10.2 apply to optimisation methods and not to these.

Q-P04-09 Which results in Theorems A–C are new, and which are specialisations of established self-normalised importance-sampling and concentration theory?

Q-P04-10 Give primitive sufficient conditions for the uniform integrability assumed in Theorem 4.1(iii), or say that none is available.

Q-P04-11 Why does minimising the empirical surrogate action not minimise the asymptotic variance, the chi-square divergence or the truth-link error? Your own example separates the two minimisers by a factor of four in variance.

Q-P04-12 What exactly is estimated when M, λ, the proposal, or the discretisation convention changes? Is any estimand invariant across nested grids, and can the family $\{Q_{\nu,M}\}$ be made projectively consistent?

Q-P04-13 Supply a second worked example with non-constant drift divergence, so that the Itô term is exercised and Tier I is tested in the regime for which it was written.

Q-P04-14 Raise the replication count of the convergence study, archive the seeds and code, and report the spread of the variance statistic; at fifty replications its batch-to-batch variability exceeds the effect described in the caption.

Q-P04-15 How would the theory change for a state-dependent diffusion $\sigma(x)$, where the classical action is sensitive to the coordinate system and to the discretisation convention?

Q-P04-16 What empirical procedure would allow (D1) to be tested or bounded in an application where the true path law is unknown, and what decision consumes $F^*(\nu)$ when it cannot be?

3 Questions of the portfolio-level evaluation

Source: portfolio-level technical evaluation, version 2, 4 September 2026 (Exhibit H, 02_technical_evaluations), Section 7. Question 6 concerns Paper 4; Question 9 concerns all six papers.

Q-PF-06 **W4 hypothesis (D1).** Beyond the conjectured control by $J(R)$ or the memory rank, is there any presently available route to *estimating* $\eta_{\nu,M}$ for a given proposal–surrogate pair? If not, state in the abstract that the estimand is the selected measure and not the target.

Note. Q-P04-04 and Q-P04-16 bear on the same hypothesis.

Q-PF-09 **Deposit venue.** State whether the works will be deposited on arXiv (math-ph / math.PR / quant-ph).